

Final Year Proposal Report on
Network Intrusion Detection AI



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Submitted By
Rajish Kumar Pandey(022-357)
Shristi Dhakal(022-376) Sulab
Poudel(022-379)

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Abstract

Contents

Chapter 1:Introduction.....	1
1.1. Overview	1
Chapter 2: Literature Review	6
Chapter 3: System Design	9

Chapter 1: Introduction

As internet technology develops quickly through digital transformation, the nature of cybersecurity threats has changed significantly. Many organizations rely heavily on their computer networks for communication, data storage, and conducting business activities, and most cyberattack avenues (e.g., malware, denial-of-service (DoS), phishing, and unauthorized access) indicate that this reliance makes organizations prime targets for cybercriminals. Traditional security techniques, like firewalls and signature-based intrusion detection systems (IDS), are usually no longer effective in detecting novel (also known as zero-day) attacks. Artificial Intelligence (AI), specifically Machine Learning (ML) or Deep Learning (DL) allows for more sophisticated approaches when detecting abnormal or suspicious behavior within network traffic and making predictions about possible intrusions. AI-enabled intrusion detection systems can analyze massive amounts of data, revealing hidden relationships between data points and continuing to improve the accuracy of their detections over time. The goal of this project is to develop and implement an AI-based Network Intrusion Detection System (NIDS) that identifies malicious activity (such as an attempted or successful intrusion) within a given network traffic dataset with a high level of accuracy and low false alarm rate. The National Institute of Standards and Technology (NIST) has indicated that IDSs provide a valuable component in the operation of ongoing monitoring of network activity to help identify attempts to gain unauthorized access to systems, thereby supporting system security.

1.1. Overview

A Network Intrusion Detection System (NIDS) is a cybersecurity solution designed to monitor network traffic and identify suspicious or malicious activities that may compromise system security. With the rapid increase in cyber threats such as malware, denial-of-service attacks, and unauthorized access, traditional signature-based detection methods are no longer sufficient because they struggle to detect new or unknown attacks. This project focuses on developing an intelligent intrusion detection system using Artificial Intelligence (AI) and Machine Learning (ML) techniques. The system will analyze network traffic data, learn patterns of normal and abnormal behavior, and automatically detect potential security threats with high accuracy. AI-based approaches are capable of adapting to evolving attack patterns and can significantly reduce false alarms compared to traditional systems. The proposed system will use publicly available cybersecurity datasets to train machine learning models such as Random Forest, Support Vector Machine, or Neural Networks. After training, the system will be able to classify network activity into normal or malicious categories and generate alerts when suspicious behavior is detected. A user-friendly interface will also be developed to visualize detection results and system performance. Overall, this project aims to demonstrate how artificial

intelligence can improve cybersecurity by creating a more efficient, adaptive, and accurate network intrusion detection solution suitable for modern digital environments.

1.2. Problem Statement

The rapid expansion of computer networks, cloud computing, and internet-based services has significantly increased the risk of cyberattacks and unauthorized network access. Organizations today face sophisticated threats such as distributed denial-of-service (DDoS) attacks, malware infections, phishing attempts, insider threats, and zero-day vulnerabilities. Traditional network security mechanisms, including firewalls and signature-based intrusion detection systems, primarily rely on predefined rules and known attack patterns. As a result, they often fail to detect new or evolving threats and generate a high number of false alarms, which can overwhelm security administrators and reduce system reliability. Furthermore, the increasing volume and complexity of network traffic make manual monitoring impractical and inefficient. Artificial Intelligence (AI) and Machine Learning (ML) provide promising solutions to these challenges by enabling automated analysis, anomaly detection, and adaptive learning from network data. However, implementing an effective AI-based intrusion detection system remains challenging due to issues such as dataset imbalance, feature selection complexity, computational requirements, and the need for real-time performance. Therefore, there is a strong need to design and develop an intelligent network intrusion detection system that can accurately identify both known and unknown attacks, minimize false positives, and provide timely alerts for preventive action. According to the National Institute of Standards and Technology (NIST), intrusion detection plays a critical role in protecting network infrastructure by monitoring activities and identifying potential security violations, emphasizing the importance of advanced detection technologies in modern cybersecurity environments.

1.3. Aims and objectives

1.3.1. Aims

The primary aim of this project is to design and develop an Artificial Intelligencebased Network Intrusion Detection System capable of detecting malicious activities in network traffic with high accuracy, efficiency, and reliability while minimizing false alarm rates.

1.3.2. Objectives

- To study and analyze different types of cyberattacks and intrusion detection techniques.
- To collect and preprocess network traffic datasets for training AI models.
- To implement machine learning and/or deep learning algorithms for intrusion detection.
- To train and evaluate the developed models using performance metrics such as accuracy, precision, recall, and F1-score.

- To design a prototype system capable of identifying abnormal network behavior in real time or near real time.
- To develop a user interface or visualization dashboard for monitoring alerts and results.
- To compare the performance of different AI models and select the most effective approach

1.4. Motivation

The motivation behind this project arises from the increasing number and sophistication of cyber threats affecting individuals, organizations, and governments worldwide. Cyberattacks can cause financial losses, data breaches, service disruptions, and damage to organizational reputation. Traditional intrusion detection systems are limited in their ability to detect emerging threats, especially zero-day attacks that do not match known signatures. Artificial Intelligence offers the ability to learn from historical data, detect hidden patterns, and adapt to new attack behaviors, making it a powerful tool for enhancing cybersecurity. Additionally, the growing dependence on digital infrastructure in sectors such as banking, healthcare, education, and government services highlights the need for intelligent and automated security systems. This project provides an opportunity to apply AI concepts to real-world cybersecurity problems and contribute to the development of more secure network environments.

1.5. Scope and Application

1.5.1. Scope

The scope of this project includes the design, development, and evaluation of an AI-based network intrusion detection system using publicly available datasets. The system will focus on detecting various types of network attacks such as denial-of-service, brute force, probing, and unauthorized access attempts. Machine learning models will be trained and tested using historical network traffic data to classify activities as normal or malicious. The project will cover data preprocessing, feature engineering, model training, performance evaluation, and visualization of results. However, the system may be implemented as a prototype or simulation rather than a fully deployed enterprise-level security solution. Real-time deployment in large-scale production environments and integration with commercial security infrastructure may be considered as future work.

1.5.2. Application

The proposed AI-based network intrusion detection system can be applied in multiple domains where network security is critical:

- **Enterprise Network Security:** Organizations can use the system to monitor internal network traffic and detect unauthorized access or malicious activities.
- **Banking and Financial Institutions:** Financial systems require strong security to protect sensitive data and transactions from cyber threats.

- **Cloud Computing Environments:** Cloud service providers can implement AI-based IDS to secure virtual networks and cloud infrastructure.
- **Government and Defense Systems:** Government agencies can use intrusion detection systems to protect confidential information and national infrastructure.
- **Healthcare Systems:** Hospitals and healthcare organizations can secure patient data and medical devices from cyberattacks.

1.6. Feasibility Study

Feasibility analysis evaluates whether the proposed AI-based Network Intrusion Detection System (NIDS) can be successfully developed and implemented within available constraints such as time, cost, and resources.

• Technical Feasibility

The project is technically feasible because the required technologies, tools, and datasets are readily available. Programming languages such as Python provide extensive libraries including Scikit-learn, TensorFlow, and PyTorch for machine learning implementation. Publicly available datasets such as CICIDS2017, NSLKDD, and UNSW-NB15 provide labeled network traffic suitable for training and testing intrusion detection models.

Modern computers with moderate specifications are sufficient for training machine learning models. Cloud platforms such as Google Colab can also be used to access GPU resources if needed. According to the National Institute of Standards and Technology (NIST), machine learning techniques are increasingly practical for intrusion detection because of advancements in computational power and data availability.

• Economic Feasibility

The system is economically feasible because it primarily relies on open-source software and publicly available datasets.

Estimated cost factors:

- Software tools: Free (Python, libraries, Jupyter Notebook)
- Hardware: Existing personal computer
- Internet access: Minimal cost
- Development cost: Academic project level

Therefore, the project does not require significant financial investment.

• Operational Feasibility

The system is user-friendly and can be operated by network administrators with basic training. The interface will display alerts and visualization dashboards, making it practical for real-world monitoring environments. AI-based intrusion detection systems reduce manual workload by automating threat detection and analysis, improving operational efficiency.

- **Schedule Feasibility**

The project can be completed within an academic semester (approximately 4 months). Tasks such as data preprocessing, model training, and system development are manageable within the given timeframe.

Chapter 2: Literature Review

2.1. Introduction

NIDS is a security system intended to analyze network traffic and detect malicious/suspicious behavior, unauthorized access, or policy violations. Conventional security systems, like firewalls, are preventive in nature but lack the ability to detect complex or insider attacks. Thus, IDSs have become a crucial part of contemporary security systems.

The initial IDS systems were mainly based on rule-driven detection methods, which compared traffic patterns with known attack patterns. However, with the rise in complexity of networks and the advent of zero-day attacks, these systems became inadequate, leading to the development of more advanced intelligent intrusion detection systems that use adaptive learning and pattern recognition. These developments include the use of machine learning, which provides the capability for adaptive learning, and deep learning, which improves the capacity for the detection of complex patterns, as well as hybrid models that use a combination of various methods for better results. Current research efforts are aimed at improving the accuracy of detection, minimizing false positives, and achieving real-time detection in highspeed networks.

2.2. Overview of Existing IDS

IDSs can be broadly categorized into two types: Signature-based detection, where the network traffic is matched against a database of known attack signatures, and Anomalybased detection, where the system tries to identify abnormal activity patterns. Signaturebased IDS compares network traffic with predefined attack patterns. One of the first and most successful systems developed is Snort, which proved the concept of rule-based detection in practical networks [1]. Although the accuracy of signature-based systems is very high for known attacks, they are inefficient against unknown or dynamic threats. Anomaly-based IDS models try to identify abnormal activity patterns by modeling normal network behavior using statistical and machine learning algorithms. A data mining approach for intrusion detection was also proposed by researchers, which tries to analyze audit trails to detect abnormal activity patterns [2]. This method is capable of detecting new attacks but may have higher false positives.

2.3. Machine Learning-Based IDM

Machine learning algorithms, such as supervised learning algorithms Decision Trees and Random Forest classifiers, have improved the performance of IDS systems considerably by allowing automated pattern recognition in network traffic data, resulting in improved detection accuracy and noise robustness. Supervised learning algorithms such as Decision Trees, Support Vector Machines, and Random Forest classifiers have been extensively used in intrusion detection systems. These algorithms learn from labeled datasets such as KDD Cup 1999 and NSL-KDD to distinguish between normal and malicious network traffic. Random forest-based IDS systems have shown high accuracy and noise robustness [3]. Unsupervised learning techniques such as clustering and density-based detection

techniques are employed when labeled attack data is not available. These techniques create a baseline for normal network activity and detect anomalies based on the baseline [4].

2.4. Deep Learning-Based Intrusion Detection

Recent studies have concentrated on applying deep learning methods, including Recurrent Neural Networks and LSTM networks. These methods are especially useful for analyzing sequential data and detecting temporal patterns of attacks because of their capability to learn hierarchical representations of network traffic. Recurrent neural networks and LSTM networks are especially useful for analyzing sequential network data and detecting temporal patterns of attacks. It was shown that IDS models based on LSTM outperform traditional machine learning methods in complex attack detection [5]. Auto-encoders and deep belief networks were also used for unsupervised intrusion detection by learning compact representations of normal network traffic and detecting anomalies as potential intrusions [6].

2.5. Benefits of Network Intrusion Detection Models

The implementation of advanced IDS models provides several advantages:

- Early threat detection: Malicious activities are detected before compromising the system
- Automated monitoring: The need for security analysis is reduced
- Adaptive learning: Machine learning models adjust to new attack patterns
- Real-time analysis: Continuous monitoring of network traffic is possible
- Improved security posture: Internal and external threats are better protected against

The above-mentioned benefits of IDS make it a crucial part of any modern enterprise or institutional network.

2.6. Limitations and Challenges Identified in Literature

Although a lot of progress has been made, there are still some challenges that exist in the field of intrusion detection research. There are many models of IDS that have a high false positive rate, which makes it difficult to trust automated systems for intrusion detection. In addition, machine learning models are very sensitive to the quality and relevance of their training data. The typical benchmark datasets used are also outdated and may not reflect contemporary network traffic patterns. Another challenge is the issue of computational complexity, especially when using deep learning-based IDS models in realtime systems. In addition, the threat of adversarial attacks on machine learning models is a new challenge to the reliability of IDS systems [7].

2.7. Scalability and Performance Considerations

Modern networks generate massive volumes of traffic, requiring IDS models capable of high-speed analysis. Scalable IDS architectures incorporate distributed processing, cloud

deployment, and parallel computation techniques. Feature selection and dimensionality reduction methods like PCA help increase the efficiency of the system by minimizing processing costs. GPU and FPGA-based designs have also been investigated for their potential to enable real-time detection in high-speed networks

2.8. Gaps in Existing Research

The literature points out some research gaps that trigger further improvement of network intrusion detection models, such as the lack of real-world network traffic environment assessments, which are essential for the validation of IDS efficiency, and the lack of attention to lightweight IDS models for resource-constrained systems, which is significant for the extension of IDS application areas.

- Lack of assessment with real-world network traffic environment
- Lack of attention to lightweight IDS models for resource-constrained systems
- Lack of standardized datasets for contemporary cyber threats
- Lack of research on adversarial robustness of machine learning-based IDS
- Need for hybrid models that integrate signature and anomaly-based detection strategies

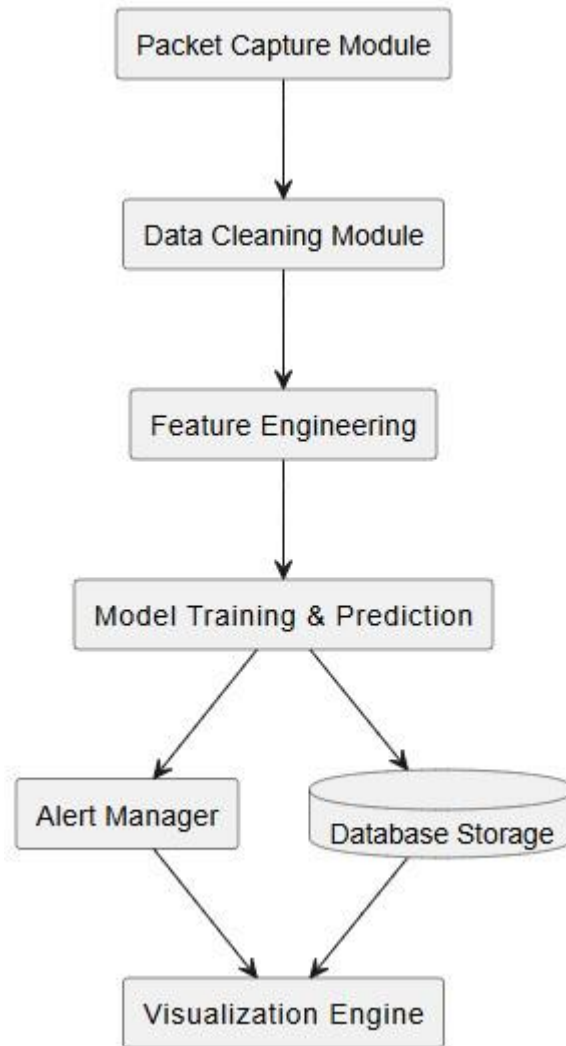
Chapter 3: System Design

The system design of the Artificial Intelligence–based Network Intrusion Detection System (NIDS) focuses on creating a structured and modular architecture that efficiently captures, processes, analyzes, and monitors network traffic to detect malicious activities. The system begins with a data acquisition module that collects network traffic data from datasets or live network sources, followed by a preprocessing module responsible for cleaning, normalizing, and transforming the data into a suitable format for analysis. A feature engineering component then selects the most relevant attributes to improve model performance and reduce computational complexity. The processed data is passed to the machine learning–based detection engine, where trained algorithms classify network behavior as normal or malicious. Once an intrusion is detected, the alert management module generates notifications and stores logs in a database for further analysis. Finally, a visualization dashboard provides users with real-time monitoring, reports, and performance metrics through an interactive interface. This layered design ensures scalability, accuracy, and efficient threat detection, aligning with cybersecurity recommendations from organizations such as the National Institute of Standards and Technology, which emphasize continuous monitoring and automated analysis for effective network protection.

3.1. System Architecture

Figure 1 : System

The system's internal sequential, modular raw network signals Initial traffic by rigorous data optimization to learning models inputs for detection engine breach, the system to response by alerts and logging the forensics. This culminates in a interface, allowing monitor network AI-driven insights scalable process.



3.2.Activity Diagram

The system's workflow is represented by a sequence of interconnected operations that govern the transition from data acquisition to threat neutralization. Upon initialization, the system enters a continuous monitoring loop where incoming traffic is ingested, sanitized, and normalized for algorithmic compatibility. The core logic resides in a central decision fork: the machine learning model performs real-time inference to evaluate the risk profile of the traffic. If the behavior aligns with known attack patterns or statistical anomalies, the process branches into an active alert state, triggering notifications and defensive logging. In the absence of a threat, the system seamlessly recycles back to the capture phase. This diagram effectively illustrates the system's "sense-think-act" cycle, providing a clear visual guide for the internal execution logic and event-driven triggers.

Architecture

structure is built on a framework that converts into defensive actions. acquisition is followed sanitization and feature ensure the machine receive high-quality classification. Once the confirms a security transitions from analysis generating localized incident data for future granular architecture centralized visualization security teams to health and respond to through a streamlined,

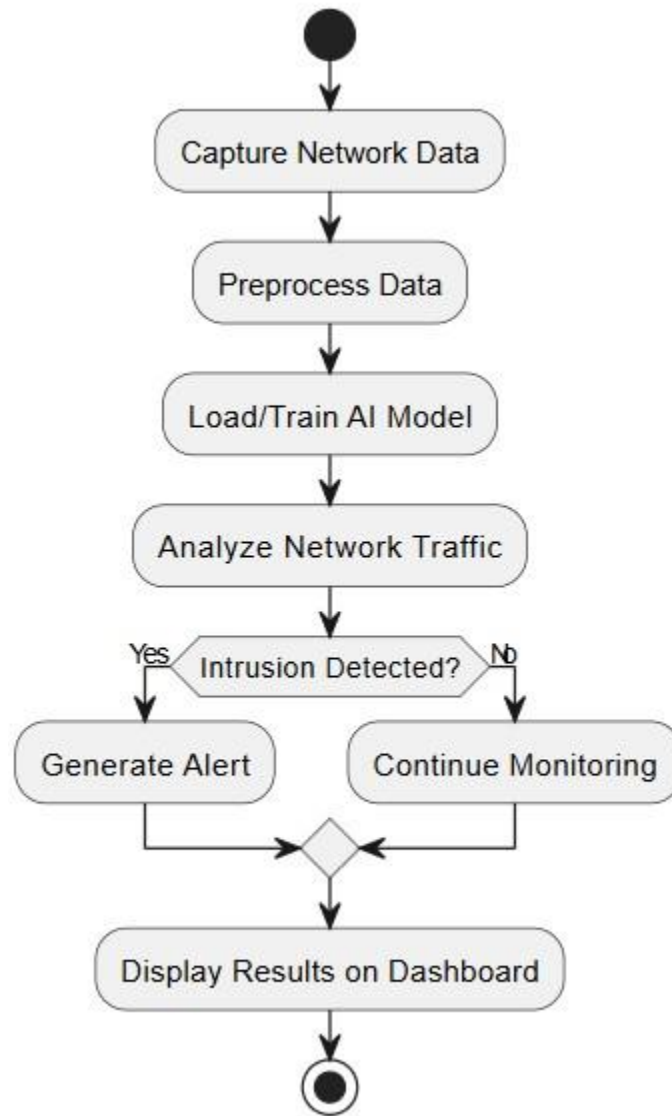


Figure 2 : Activity Diagram

3.3.Data flow diagram

The Level 1 Data Flow Diagram provides a detailed functional breakdown of the NIDS, illustrating the transformation of raw network packets into actionable security intelligence. By expanding the high-level system into discrete operational modules, this diagram highlights the specific interactions between internal processes, external entities (such as network interfaces and administrators), and persistent data repositories. In this AI-driven framework, the Level 1 DFD tracks the lifecycle of data as it is captured from the stream, sanitized during preprocessing, evaluated by the machine learning engine, and finally archived as historical logs or broadcasted as real-time visual alerts.

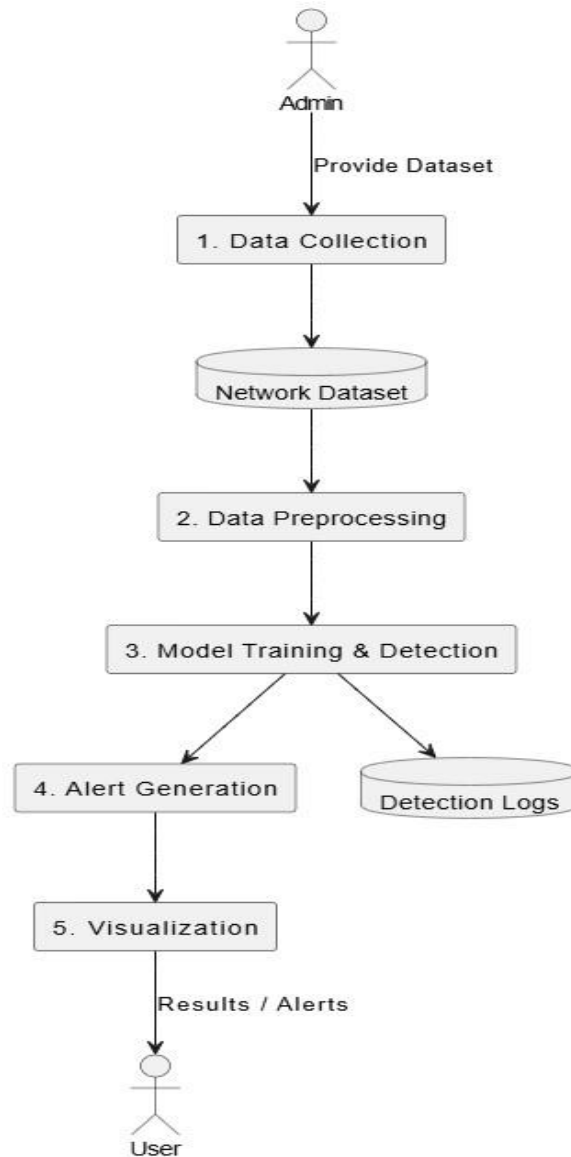


Figure 3 : Data Flow Diagram

1. External Entities

The Data Flow Model indicates that the machine learning system is a conduit between the automated logic (in this case, machine learning) of an algorithm and its users (i.e., human operators). In general, there are two main external parties that access or interact with a machine learning platform:

- 1.1. **Administrator (System Orchestrator)** :The Administrator interacts with the system's backend and configuration pipelines. Their primary role is to ensure the operational integrity and accuracy of the AI model. In the context of the data flow, the Administrator:
 - **Supplies Data:** Ingests the initial training datasets (e.g., NF-UQ-NIDS-v2) into the preprocessing module.

- **Manages the Model:** Configures hyperparameters, initiates model training/retraining phases, and updates the detection engine when network behaviors drift.
 - **Configures the Environment:** Defines the specific network interfaces for packet sniffing and sets the sensitivity thresholds for the alert generation module.
- 1.2. **User / Security Analyst (Threat Responder):** The Security Analyst interacts with the system's frontend and output pipelines. Their role is heavily consumer-oriented, focusing on incident response rather than system configuration. In the context of the data flow, the Analyst:
- **Consumes Telemetry:** Monitors the real-time visualization dashboard to assess overall network health.
 - **Triages Alerts:** Receives high-priority notifications generated by the Intrusion Detection Engine to validate whether an anomaly is a true threat or a benign event.
 - **Conducts Forensics:** Queries the historical database to analyze intrusion reports, track the origin of flagged IP addresses, and adjust organizational security policies based on the AI's findings.

Chapter 4: Conclusion

The proposed Artificial Intelligence–based Network Intrusion Detection System (NIDS) provides an efficient and intelligent solution for improving network security by automatically analyzing network traffic and detecting malicious activities with high accuracy and reduced false alarms. By combining machine learning techniques with structured modules such as data collection, preprocessing, feature engineering, detection, alert generation, and visualization, the system ensures scalability, adaptability, and reliable performance in modern network environments. The feasibility study and system design demonstrate that the project can be successfully implemented using available tools and datasets within practical constraints, while also offering opportunities for future enhancements such as real-time deployment and cloud integration. Overall, this project highlights the importance of AI-driven cybersecurity solutions

in protecting digital infrastructure, aligning with best practices recommended by the National Institute of Standards and Technology for continuous monitoring and advanced threat detection.

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